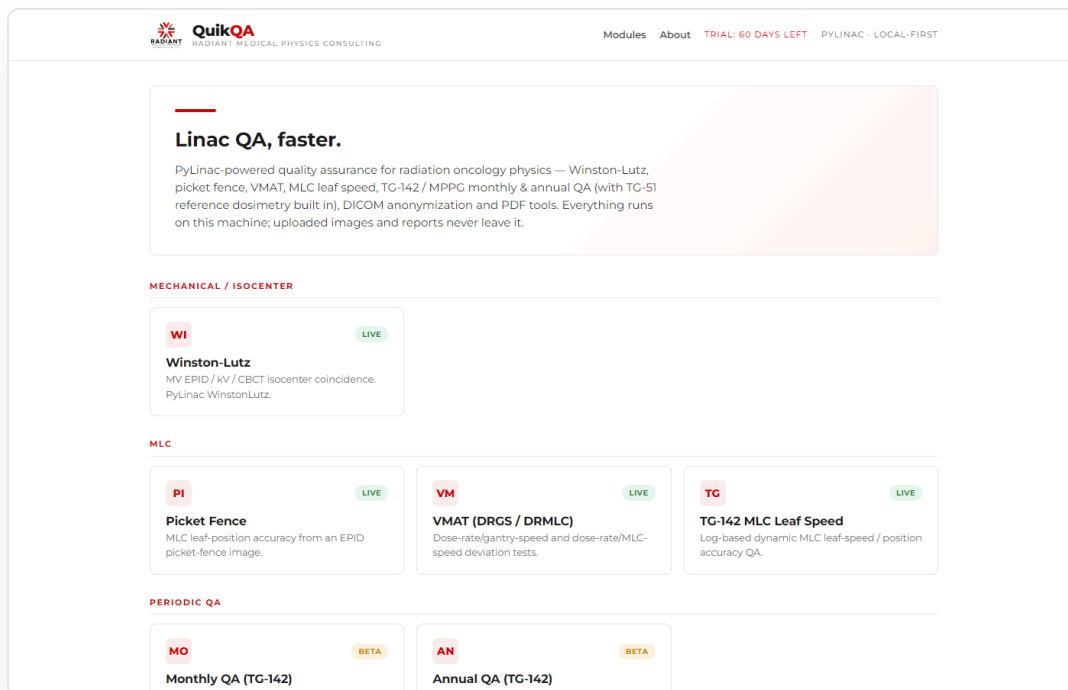




USERS GUIDE

Radiant QuikQA

PyLinac-powered TG-142 / MPPG linac QA in one local web app.



What QuikQA is.

Radiant QuikQA is PyLinac-powered machine quality assurance for radiation oncology physics — the full TG-142 and MPPG 8.b / 9.a program in one local web app, with guided Monthly and Annual worksheets backed by automated Winston-Lutz, Picket Fence, VMAT, and MLC leaf-speed analysis, plus a TG-51 reference-dosimetry calculator.

QuikQA was built for the clinical physicist who runs linear-accelerator QA on a busy clinical schedule. It consolidates the worksheets, the tolerance tables, the analysis engines, and the report layout into one tool — so the entire monthly visit, or the full annual program, lives in the browser on the QA workstation.

Who it's for

- Clinical and QA medical physicists running TG-142 monthly and annual programs.
- Linac commissioning teams.
- Physics residents and students learning the TG-142 / MPPG framework.

Evaluation build. QuikQA is an unvalidated analysis aid. All tolerances and outputs must be independently reviewed and approved by a Qualified Medical Physicist before operational use.

Linac QA, faster.

PyLinac-powered quality assurance for radiation oncology physics — Winston-Lutz, picket fence, VMAT, MLC leaf speed, TG-142 / MPPG monthly & annual QA (with TG-51 reference dosimetry built in), DICOM anonymization and PDF tools. Everything runs on this machine; uploaded images and reports never leave it.

MECHANICAL / ISOCENTER

WI

LIVE

Winston-Lutz

MV EPID / kV / CBCT isocenter coincidence.
PyLinac WinstonLutz.

MLC

PI

LIVE

Picket Fence

MLC leaf-position accuracy from an EPID picket-fence image.

VM

LIVE

VMAT (DRGS / DRMLC)

Dose-rate/gantry-speed and dose-rate/MLC-speed deviation tests.

TG

LIVE

TG-142 MLC Leaf Speed

Log-based dynamic MLC leaf-speed / position accuracy QA.

PERIODIC QA

MO

BETA

Monthly QA (TG-142)

Full TG-142 / MPPG 8 hr monthly QA checklist.

AN

BETA

Annual QA (TG-142)

Full TG-142 / MPPG 9 annual QA checklist.

MODULE HUB — PICK A QA WORKFLOW.

Access and license activation.

Open the app

QuikQA is deployed as a local-feeling web app at quikqa.radiant-mpc.com. Open the URL in any modern browser — no Python install, no ESAPI dependency, no software to deploy on the QA workstation.

Sign in & activate your license

On first visit, QuikQA displays a license-activation modal. Paste the RADKEY string from your Radiant license email and click *Activate*. The key is stored locally in your browser; subsequent visits are gated transparently.

License key format	RADKEY-XXXXX-XXXXX- ...
Activation	Online verification against the Radiant signing service
Expiration	Per the term in your agreement (30 / 60 / 365 days typical)

What's on the hub

The hub lists every module as a card: TG-142 Monthly, TG-142 Annual, Winston-Lutz, Picket Fence, VMAT, MLC Leaf-Speed, TG-51 Photon, TG-51 Electron, plus data-tool cards for the DICOM Anonymizer and the PDF toolkit. Click a card to enter that module.

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Full TG-142 / MPPG 9 annual QA checklist.

QUIKQA HUB — TG-142 MODULES, ANALYSIS ENGINES, DATA TOOLS.

TG-142 Monthly QA worksheet.

The Monthly worksheet is the heart of the routine QA visit — the complete TG-142 catalog with MPPG 8.b tolerances, switchable between IMRT and SRS class.

1. **Header.** Enter the machine, performer, and energies delivered. The header is reused on every page of the rendered report.
2. **Dosimetry.** Type the measured deviation for each tested item (output constancy, profile, backup monitor, etc.). QuikQA grades each row against the tolerance live, with green / amber / red indicators.
3. **Mechanical & safety checks.** Set each functional check to PASS / ACTION / FAIL with one click.
4. **Imaging.** MV, kV, and CBCT isocenter and scaling rows complete the catalog.
5. **Submit.** QuikQA computes a worst-of roll-up per group and overall, and renders a print-ready, Radiant-branded PDF report.


QuikQA
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[PYLINAC - LOCAL-FIRST](#)

[Modules / Monthly QA \(TG-142\)](#)

Monthly QA (TG-142)

Full TG-142 / MPPG 8.b monthly QA checklist with tolerance grading and a printable report.

Record each test result below. For measured tests, enter the **deviation** from the expected value — a result passes when its absolute deviation is within the tolerance shown. Tolerances are pre-filled with the selected machine class default and may be edited per test.

MONTHLY QA — HEADER

Machine / linac name

QA date

Performed by

Machine class — sets default tolerances

IMRT Only

Energies / notes (optional)

DOSIMETRY

Test	Tolerance	Result
X-ray output constancy (all photon energies) — TG-142 Table II	± 2.0 %	deviatik %
Electron output constancy (all electron energies) — TG-142 Table II	± 2.0 %	deviatik %
Backup monitor chamber constancy — TG-142 Table II	± 2.0 %	deviatik %

TG-142 MONTHLY FORM — HEADER AND TOLERANCE-GRADED ROWS.

TG-142 Annual QA + TG-51 reference dosimetry.

The Annual worksheet is the same flow at annual cadence, with an embedded TG-51 launcher panel. From inside the Annual form you can open either the TG-51 Photon or TG-51 Electron calculator, complete the absolute output cal, transcribe the result back into the row, and submit the consolidated annual report.

TG-51 calculator

- Built-in chamber and electrometer catalog with ADCL fields.
- Pressure / temperature, P_{ion} , P_{pol} , P_{elec} , k_Q corrections.
- One-click round-trip back to the Annual worksheet.

Independent verification. TG-51 outputs from QuikQA must be cross-checked against an independent calculation before use as a calibration result.


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Machine / linac name

QA date

Performed by

Machine class — sets default tolerances

Energies / notes (optional)

DOSIMETRY

TG-51 reference dosimetry. Run the absolute output calibration in the dedicated TG-51 calculator for each photon and electron energy, then record the result below on "X-ray / electron output calibration, TG-51" as the deviation from your expected output. Each calculator opens in a new tab, so this checklist is preserved.

Open TG-51 Photon calculator ↗
Open TG-51 Electron calculator ↗

Test	Tolerance	Result
X-ray / electron output calibration, TG-51 (absolute) — TG-142 Table III	± 1.0 %	deviati %
Photon flatness change from baseline — TG-142 Table III	± 1.0 %	deviati %
Photon symmetry change from baseline — TG-142 Table III	± 1.0 %	deviati %
Electron flatness change from baseline — TG-142 Table III	± 1.0 %	deviati %
Electron symmetry change from baseline — TG-142 Table III	± 1.0 %	deviati %
Photon beam quality (PDD10 or TMR20/10) from		

TG-142 ANNUAL — TG-51 LAUNCHER PANEL.

Winston-Lutz, Picket Fence, VMAT, MLC leaf-speed.

Every analysis module is PyLinac under the hood, wrapped in a guided form with explicit machine profile, tolerance, and upload steps. Each returns a metrics table, the analysis plot, and a downloadable PDF.

Winston-Lutz

Drop a full MV, kV, or CBCT image set. QuikQA computes isocenter, BB-to-radiation coincidence, and per-axis errors with sub-millimeter reporting.

Picket Fence

MLC leaf positional accuracy for the TG-142 monthly check, with clear pass/fail grading.

VMAT (DRGS / DRMLC)

Dose-rate and MLC-speed constancy analysis from a single delivery.

MLC leaf-speed

Log-based per-leaf RMS error, beam holds, and a combined batch report — no portal dosimetry required.



Modules / Winston-Lutz

Winston-Lutz

MV EPID / kV / CBCT isocenter coincidence. PyLinac WinstonLutz.

Upload the complete Winston-Lutz image set (MV EPID, kV, or CBCT). You may select multiple DICOM files or a single ZIP archive. PyLinac auto-detects CBCT-derived datasets.

IMAGES

WL DICOM images — select all gantry/collimator/couch images, or one ZIP

[Choose Files](#) No file chosen

MACHINE & TOLERANCE

Machine profile

TrueBeam Millennium 120

BB diameter (mm) — blank = profile default (5 mm)

5.0

OPTIONS

- ☐ Use SRS / SBRT tolerances (tighter than routine TG-142)
- ☐ Open-field analysis (no MLC-defined field)
- ☐ Low-density BB (kV / CBCT phantom)
- ☐ Apply virtual corrective shift to BB

Report notes (optional)

Run analysis

Cancel

WINSTON-LUTZ MODULE — UPLOAD FORM.

One PDF per session.

Every module — worksheet or analysis — produces a Radiant-branded PDF report you can save to the chart, the QA binder, or email to a reviewer.

What's in the report

- Header (machine, performer, date, energies).
- Per-row deviation, tolerance, and PASS / ACTION / FAIL verdict.
- Worst-of roll-up per group and overall.
- Analysis plots (Winston-Lutz, Picket Fence, VMAT, leaf-speed) inline.
- Reviewer signature block on the last page.

Data tools

- **DICOM Anonymizer** — strip identifying tags before sharing test data.
- **PDF toolkit** — merge, split, rotate, watermark, and number QA reports.

**QuikQA**
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Modules About TRIAL: 60 DAYS LEFT PYLINAC : LOCAL-FIRST

Modules / Picket Fence

Picket Fence

MLC leaf-position accuracy from an EPID picket-fence image.

Upload a single picket-fence EPID DICOM image. An optional Varian trajectory log enables absolute (vs. fitted-line) comparison — recommended for TrueBeams.

IMAGE & LOG

Picket-fence DICOM image

Choose File No file chosen

Trajectory log (optional — .bin / .dlog)

Choose File No file chosen

MACHINE & TOLERANCE

Machine profile

TrueBeam Millennium 120

Tolerance (mm) blank = 0.5

0.5

Action level (mm) blank = 0.35

0.35

Sag adjustment (mm)

0.0

Crop margin (mm)

3

Report notes (optional)

PICKET FENCE MODULE — ANALYSIS FORM.

License, safety, and how to reach us.

License terms

QuikQA is licensed per workstation, per seat, for the term in your agreement. The license string is verified at activation and validated against a local clock-guard to catch backdated systems.

Evaluation status

QuikQA is distributed as an *evaluation build*. It is not a medical device. It is an analysis aid for a Qualified Medical Physicist. Outputs must be reviewed before any clinical, dosimetric, or release-to-service decision.

Privacy

QuikQA runs entirely on your machine. Uploads (DICOM images, log files) and outputs (PDFs) live in the workstation's filesystem and are not transmitted off the box. Do not upload PHI.

Contact

Live URL: <https://quikqa.radiant-mpc.com/>

Email: info@Radiant-MPC.com

Website: radiant-mpc.com

Evaluation / decision-support only. Every Radiant tool is an aid to a Qualified Medical Physicist. All outputs must be independently reviewed before any clinical use. Do not transmit PHI through evaluation builds.